Group Research

Analysis of Factors Impacting Export Performance in Sri Lanka

by

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DECLARATION

We, the undergraduate students of the University of Sri Jayewardenepura, hereby declare that the research project titled "Analysis of the Factors Impacting Export Performance in Sri Lanka" has been carried out by us under the supervision of Dr. Neluka Devpura, Senior Lecturer, Department of Statistics, Faculty of Applied Sciences, University of Sri Jayewardenepura. We hereby declare that this research is an original work completed by our group as part of the requirements for the STA 351 2.0 Research Methodology. We further declare that this research, in whole or in part, has not been submitted to any other University or academic institution for any degree, diploma, or certificate. We take full responsibility for the content, analysis, and findings presented in this report.

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ABSTRACT

Exports play a fundamental role in economic growth and development, particularly for small open economies like Sri Lanka. Since the liberalization reforms of 1977, Sri Lanka's economic history has shown a consistent effort to engage with the global economy through the encouragement of export-driven growth. However, despite these efforts Sri Lanka's export performance during the past seventeen years has been quite unstable. Therefore, the present study attempts to examine how imports, exchange rate and crude oil price affect Sri Lanka's export performance using monthly data over the period from 2007 to 2024. The study uses first differences of logarithmic transformations of all the variables and uses multiple linear regression. This model allows the interpretation of coefficients as short-run elasticities representing the percentage change in exports associated with a 1% change in each predictor variable. The findings of the study reveal that in short run, import growth positively and significantly affects export growth. Domestic currency depreciation (positive growth in exchange rate) negatively affects exports but this relationship is statistically insignificant. Similarly, crude oil price growth has a positive but statistically insignificant effect on export growth. Hence, in this model, exports are only significantly impacted by imports. The policy implications suggest that measures designed to restrict imports may unintentionally harm export competitiveness. Instead of broad restrictions, policymakers should ensure the smooth flow of essential imports while discouraging unnecessary luxury imports that drain foreign reserves. Although insignificant, the relationship between exports and crude oil price highlights Sri Lanka's vulnerability to external shocks and underline the need for export diversification and the reforms in the energy sector. The negative relationship between exports and imports, though insignificant, suggests that policymakers should not rely solely on currency depreciation as a tool for promoting exports. Instead, they should aim for exchange rate stability and focus on enhancing production efficiency, diversification, and cost reduction.

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CHAPTER 1
Introduction

1.1 Motivation

Exports play a vital role in a country's economy by increasing national income, creating employment, improving technological capabilities, encouraging for investments and expanding international relationships. Since the liberalization reforms of 1977, Sri Lanka's economic history has shown a consistent effort to engage with the global economy through the encouragement of export-driven growth. Traditional export sectors like tea, rubber, and coconut initially led the country's trade performance; however, over the years, manufactured exports, especially apparel and textiles, became the foundation of the export economy. In recent years, sectors like information technology, tourism, and business process outsourcing have begun to add to the nation's external earnings.

Sri Lanka's export performance during the past seventeen years has been quite unstable in spite of these efforts. The percentage of exports in GDP has steadily fallen relative to other regional economies, showcasing structural weaknesses and external vulnerabilities. For instance, although clothing exports are a significant factor, they are very responsive to fluctuations in global demand and changes in exchange rates. In the same way, agricultural exports like tea and rubber encounter difficulties due to climate change, variable global commodity prices, and rising competition from other exporting countries. Concerns have been raised concerning Sri Lanka's export industry's durability and long-term ability to support sustainable economic growth as a result of this uneven performance.

Exports continue to be an essential engine for Sri Lanka's economy; however, the nation's export to GDP ratio has consistently decreased in recent years, falling behind regional competitors like Vietnam and Bangladesh. This ongoing underperformance, even with consecutive government policies like the National Export Strategy, has raised important inquiries into the true factors influencing export growth. Moreover, although numerous studies have investigated the impact of exchange rates or global demand on exports in different economies, there is limited empirical evidence regarding how macroeconomic variables like imports and global oil price changes affect Sri Lanka's export performance. This gap in both the literature and policy analysis provides a strong motivation for the present study, which aims to produce context specific insights that can inform trade and economic policy.

The success of exports is influenced by a range of interconnected domestic and external variables. Imports, exchange rates, and global crude oil prices are notably important in Sri Lanka's situation. Imports, particularly of intermediate goods and raw materials, are an essential for industries focused on exports. For example, the clothing industry depends significantly on foreign textiles and equipment, whereas agricultural exports rely on imported fertilizers and fuel. As a result, imports frequently function as a facilitator of exports rather than as a competitor. At the same time, fluctuations in exchange rates directly influence the competitiveness of exports. A decline in the value of the Sri Lankan rupee could improve the price competitiveness of exports in global markets however, high volatility can induce uncertainty for exporters, particularly in an economy that heavily depends on imported resources.

Global crude oil prices have an indirect yet significant impact. As a net oil importing nation, Sri Lanka experiences increased production and transportation expenses when global oil prices increase, impacting the competitiveness of its export products. Additionally, fluctuations in oil prices contribute to inflationary pressures, instability in exchange rates, and challenges in the balance of payments, all of which impact the export sector. Given these interconnections, it is crucial to evaluate how these essential economic factors; imports, currency exchange rates, and crude oil prices influence the export performance of Sri Lanka.

1.2 Research problem

Export performance is among the key determinants of Sri Lanka's economic growth. However, Sri Lanka's export performance has shown a fluctuating trend in the past decade. This study aims to identify how the key economic factors; imports, exchange rate, and global crude oil prices have affected Sri Lanka's exports over the period 2007-2024.

1.3 Objectives and Aims

The key issue examined in this study is the variable nature of Sri Lanka's export performance over the last seventeen years. Even with government initiatives like the National Export Strategy and attempts at diversification, exports have not shown a consistent upward trend. The purpose of this study is to explore the impact of imports, changes in exchange rates, and average crude oil prices on Sri Lanka's exports from 2007 to 2024. The overall objective of the research is to determine the factors affecting Sri Lanka's export performance during this time period. The study specifically aims to analyse the effect of imports on export performance, to evaluate how exchange rate volatility impacts exports, to assess the consequences of crude oil price variations, and to identify which of these factors has the most substantial effect on Sri Lanka's export sector. Based on these objectives, the study is guided by the following research questions:

1. How do imports influence Sri Lanka's export performance?
2. How do the changes in Exchange rate impact Sri Lanka's exports performance?
3. How do changes in global crude oil prices affect Sri Lanka's exports?
4. Which of these factors exerts the most significant influence on Sri Lanka's export performance?

1.4 Scope of the study

The research focuses on the timeframe from January 2007 to December 2024, utilizing monthly secondary data obtained from the Central Bank of Sri Lanka, World Bank and the Federal Reserve Bank of St. Louis. The dependent variable signifies export value calculated in US dollars, whereas the independent variables include imports, the exchange rate (LKR/USD), and average worldwide crude oil prices (US$ per barrel). Due to data limitations and the need to keep the focus on the most important macroeconomic variables, other factors such as domestic political conditions, international trade agreements, and trends in global demand that may also have an impact on export performance are not considered within the scope of this study.

1.5 Significance of the study

The significance of this study is found in its practical and academic contributions. The results will offer evidence driven insights for policymakers to develop trade, economic, and exchange rate strategies that promote consistent and competitive export expansion. A better comprehension of imports' role could also aid in strategies that more effectively integrate global value chains. From an academic viewpoint, the research adds to the current literature on export performance by integrating various macroeconomic factors within the Sri Lankan context, where empirical data is relatively limited compared to developed economies. The study enhances policy discussions on how external factors like oil price fluctuations interact with internal weaknesses to influence export results.

1.6 Limitations of the Study

This study faces several limitations:

1. **Variable Limitation**: The analysis is confined to three macroeconomic factors (imports, exchange rates, crude oil prices), leaving out other potentially influential variables such as political instability, institutional quality, trade agreements, and global demand shocks.
2. **Data Limitation**: The study relies solely on secondary data from international and national databases, which may contain issues of measurement accuracy, reporting delays, or inconsistencies.
3. **Timeframe Limitation**: The study period (2007–2024) may not capture long-term structural shifts in trade or the impact of very recent policy changes that fall outside this window.
4. **Generalizability Limitation**: Findings are specific to Sri Lanka and may not be directly generalizable to other developing economies, though they may provide useful comparative insights.

1.7 Dissertation Organization

This dissertation is organized into six chapters. Chapter One introduces the study by outlining the motivation, research problem, objectives and aims, scope of the study, significance of the study, limitations of the study, and overall structure. Chapter Two reviews the existing literature on exports, imports, exchange rate dynamics, and crude oil price effects, while also identifying the research gaps that this study seeks to address. Chapter Three describes the research methodology, including the data sources, research design, hypothesis development and statistical techniques employed such as the Augmented Dickey-Fuller (ADF) test and multiple linear regression. Chapter Four presents the results of the empirical analysis, highlighting the relationships between imports, exchange rates, oil prices, and export performance, presentation of findings, statistical analysis, trends, or qualitative results. Chapter Five describes discussion including interpretation of results in context of research questions and literature, implications of findings, comparison with previous studies. Finally, Chapter Six concludes the dissertation by summarizing the key findings, contributions of the study, recommendations for practice and further research.

CHAPTER 2
Literature Review

Several studies have been reviewed to investigate the impact of the export performance of Sri Lanka. It is a widely known fact that the export performance of a country plays a major role in the economic growth of that country. Since Sri Lanka is a developing country in South Asia, it is crucial to understand the potentially significant factors for the export economy. Export influencing factors have changed in accordance with the background of the country and its economic criteria. Factors exchange rate, imports and crude oil prices have been taken into account.

The real effective exchange rate is a significant factor in considering the export performance in developed countries as Canada (Hassan et al.,2022). Naseri et al., (2021) highlight that there are 2 types of factors that influence a country's exports. They are supply-side factors and demand-side factors. Five SAARC country analysis which is, including Sri Lanka, have identified the demand side factor as the real exchange rate and the supply side factor as imports. In countries like Sri Lanka buy copious oil as a raw material for manufacturing various products. When oil prices rise, it costs more to make and ship goods, which makes exports less competitive. But when oil prices fall, it becomes cheaper to produce goods, helping exports do better. Kousar et al., (2020) studied how oil prices affect exports in Pakistan. This significantly shows that imports and crude oil are essential factors in considering Sri Lanka's exports.

2.1 Impact of imports on exports

Imports have a significant impact on export performance, particularly in open economies like Sri Lanka, where domestic production processes heavily rely on imported inputs like fuel, machinery, and raw materials (Musleh-ud Din,2004). International trade involves both exports and imports. Countries with abundant natural resources can benefit from foreign export goods and services, as well as foreign remittances, which can increase per capita income. Imports may improve income per capita in countries with limited natural resources. Innovations in manufacturing will increase per-capita incomes by exporting products and services to foreign countries (Ali et al.,2021). Adding further support, Handoyo et al.,2023 examined Indonesia's manufacturing industries across low, medium, and high technology levels. Their findings showed that raw material imports significantly boosted export performance in all categories. By applying stochastic frontier analysis, the researchers demonstrated that both technical efficiency and access to imported materials contributed to higher exports. The study also emphasized the need for policies that ease restrictions on intermediate goods imports to strengthen export-oriented production. Taken together, these studies suggest that Sri Lanka should focus on facilitating the import of essential production inputs in order to expand its export capacity. Restrictive trade policies that limit access to these goods may hinder export growth, particularly in sectors where local resources cannot fully meet production needs.

Lee et al.,2016 have emphasized that Malaysia's exports have a positive relationship with imports. The total imports and exports are positively correlated with total foreign investments (Yi,2020). In spite of that, when considering Sri Lanka, Velnampy et al., (2013) has investigated the impact of exports and imports on economic growth. A regression analysis has been performed by taking the dependent variable as economic growth and the independent variables as exports and imports. As a result of that, a positive relationship was found between imports and economic growth. A thirteen percent adjusted R-squared value indicates a 13 percent of the economic growth is affected by imports. The results obtained by the correlation analysis method have revealed that there's a strong positive correlation (98%) between exports and imports. Sri Lankan data shows that the textile-related raw materials have a greater share of export earnings. The multiple linear regression model supported the data used in Khalid (2021) claim that imported capital and intermediate goods are necessary to maintain export output by showing that imports have a positive impact on export performance. More detailed insights might have been obtained if the study had made a distinction between capital or intermediate items and consumption imports. The study pointed out that the reason behind the downturn in exports and imports is the global financial crisis. Furthermore, central bank data exhibits that exports have grown at an average annual rate of 4–6% over the past two decades, while imports have grown faster due to increasing demand for intermediate goods, fuel, and consumer products (Velnampy et al., 2013). This mismatch has resulted in constant current account deficit.

On the contrary, exports and geographical distance have a significant negative relationship in the period from 2000-2016. Sri Lanka's exports are negatively impacted by both the Indo-Sri Lanka Free Trade Agreement and SAARC Preferential Trading Arrangement (SAPTA). Due to these agreements, Sri Lanka has been able to lower prices from India, which has caused the trade balance to deteriorate heavily (Jayasinghe et al.,2022). Therefore, there is no discernible effect of the exchange rate increase on export performance.

Through monetary supply and fiscal policies could help to increase exports motivating small, medium enterprises to take an export-oriented approach by increasing the flexibility of raw material import limits (Hassana and Nufile,2016). The result of the study shows that Sri Lanka's open economy may see an increase in imports and exports under the free-floating exchange rate system. Sri Lanka depends on imports and imports cost more than the exports bring in. It caused occur budget deficit in Sri Lanka. The government of Sri Lanka must regulate imports and the country's inflationary situation in order to control this situation.

2.2 Impact of Exchange Rate on Exports

Real exchange rate is the most significant variable affecting exports (Naseri et al., 2021). However, real-world studies show mixed results, as the impact of exchange rate changes can differ based on the country and how the economy works (Jacob et al.,2021). In most of, the net value of the exchange rate is considered. But in some cases, the exchange rate relative to multiple countries is analyzed. The results of an Indian study (Gondaliya et al.,2015) show a significant causal relationship between both exports and both the Euro and the U.S dollar. In contrast, unidirectional relationships were found between exports and the pound sterling and the Japanese Yen. Those findings demonstrate the dynamic connection of India's export performance with the changes in currency.

However, by taking the net value of the exchange rate, there are several reasons for considering the relationship between exports and the exchange rate. One is when a country's currency depreciates, the goods and services of that country become cheaper for other countries, resulting in an increase in export performance. The other reason is that when the exchange rate fluctuates, it causes uncertainty for both importing and exporting countries (Ilmas et al., 2022). Iran's export performance was heavily influenced by both pricing and non-pricing factors, with exchange rate volatility playing a critical role. External shocks were found to have a negative impact on industrial goods exports, emphasizing the importance of stable exchange rates in supporting export growth in developing economies (Ebadi et al.,2015). A stable exchange rate reduces the risk to investors and exporters (Naseri et al.,2021). Except for a few years, export growth in the post-liberalization period of Sri Lanka has been positive and has exceeded 10%. According to many researchers, the real exchange rate was overvalued for many years, resulting in an anti-export bias (Dunusinghe, 2009). The Elasticity Approach to the Balance of Payments posits that exchange rate devaluation can improve a country's trade balance, provided that the price elasticity of exports and imports is sufficiently high. This is particularly relevant in Sri Lanka's case, as shown by Hassana and Nufile (2016), who observed that exports and imports increased following the liberalization of the economy and the introduction of a free-floating exchange rate system. Persistent trade deficits have historically characterized Sri Lanka's net exports. The ratio of India's export price to world export price and real effective exchange rate had a negative impact on export supply, Gurung and Rai (2020). Furthermore, according to the study, the ratio of export price to domestic price and trade openness positively affected export supply. Export demand has been considered as a function of real effective exchange rate (REER), world income, and the log of lagged export demand (Sharma,2003). REER makes Indian goods more expensive for other countries. The multiple linear regression model used in Rajakaruna (2015) emphasize that the positive relationship between exchange rate and the foreign purchases. According to the study, terms of trade is a factor of the nominal exchange rate of Sri Lanka.

According to the findings of the study, Bhavan (2016), the real effective exchange rate had a negative but significant effect on exports in the long run, but in the short run, the impact was insignificant. VECM used by Khalid et al., (2021) found the same results. In contrast, by using the same approach, short-run exchange rate effects were found significant and long run exchange rate insignificantly found in Ganeshamoorthy (2018). Johnson's co-integration approach has found that in the long run, the real effective exchange rate has a significant negative impact on exports, while currency appreciation decreases export competitiveness (Khalid,2021). A multiple linear regression method has been used to examine the factors influencing Jiangsu province's total import and export trade volume. The exchange rate is significantly negative for exports and imports (Yi,2020). In addition to that, Ahmed et al., (2017), Gururaj et al., (2016) found that the exchange rate had a negative and statistically insignificant relationship with exports. Also, the study of 5 ASEAN countries, namely

Indonesia, Singapore, Thailand, Malaysia, and the Philippines, reveals that the Exchange rate shows a significant negative relationship with export performance (Ilmas et al.,2022). Results of a multiple regression model indicate that independent variables, exports and imports affect the dependent variable exchange rate significantly. Since the Indonesian data fit the model well, the exchange rate and imports have a positive effect on the economic growth of that country (Nopiana et al.,2022).

In contradiction, Malaysia's exports have a positive relationship with imports and exchange rate by validating the Marshall-Lerner Hypothesis (Lee et al.,2016). Similarly, the Vector Error Correction Model (VECM) found that changes in the exchange rate had a positive and significant effect on India's exports (Jacob et al.,2021). Interest rates, inflation rates, and real exchange rates are all positively connected (Shimu et al.,2018). Investigations of three countries' analysis revealed that all macroeconomic variables, including interest rates, had a significant influence on exchange rates (Ramasamy et al.,2015). Regression analysis confirmed that increases in the variables exchange rate, inflation, and real interest rate significantly reduce export growth, highlighting the sector's sensitivity to macroeconomic stability (Shimu et al.,2018).

This finding goes against the usual belief that currency appreciation hurts exports, showing that the impact of exchange rate changes can depend on other factors like government policies and the structure of the economy Ilmas et al., 2022). The positive effect of the exchange rate wasn't strong enough to be considered statistically significant (Sule et al.,2021). What stood out more in their research was the importance of having a stable exchange rate. In Tanzania, (Epaphra,2016), a stable and favourable exchange rate had a strong, positive long-term impact on exports, especially when supported by trade liberalization and economic growth. Frequent fluctuations and volatility make exporters hesitant to invest for the long term because the uncertainty can impact their expected returns. (Prajanti et al.,2017) Using multiple regression analysis found that while the exchange rate and other factors together had an impact on exports, the exchange rate by itself didn't show a strong, statistically significant effect. This suggests that the influence of the exchange rate can vary depending on things like how sensitive demand is, the kinds of products being exported, and how open the economy is to trade. It is better to promote a more flexible exchange rate regime by reducing the reliance on the United States dollar (USD) (Lee et al.,2016).

Facilitating export financing facilities can lead to strengthening export performance. Kousar et al.,2022 suggest increasing both the quantity and quality of exports while lowering imports through trade restrictions like taxes and quotas. Promoting domestic goods would lead to stabilize the exchange rate and improve the trade balance. The reports also points out that as Pakistan's labour surplus has the potential to greatly boost its export industry. Exports has a negative effect on economic growth. The Indonesian government has implemented various

measures to boost export activity in order to address this issue. Through professional development, tax breaks, and integrated OSS licensing services, a number of short-term initiatives have been placed together to enhance the business environment. A rise in short-term exports, particularly in the choice of primary export products, accelerating processes to save money and time along with improving market access and economic diplomacy. The administration will also focus on the medium and long-term development of human resources and infrastructure (Nopiana et al.,2022).

2.3 Impact of Crude oil prices on exports

The link between oil prices and the economy has drawn the attention of numerous scholars since the groundbreaking study of, which found that oil prices have a negative impact on the real good. Oil price fluctuations have a significant impact on public welfare around the world (Abdelsalam,2023). As a net oil importer, rising oil prices increase domestic production and transport costs (Amarasinghe et al,2018). A strong long-run negative relationship between global oil prices and the Colombo Consumer Price Index indicates that oil price shocks fuel inflation and reduce export competitiveness indirectly. Economic policies can diminish the effect of oil price fluctuations on the actual product. The price of oil is crucial to the world economy. It remains the primary source of energy, accounting for nearly one-third of global production. As a result, economists seek to quantify how oil prices affect the real sector of the economy and the dynamics of financial markets (Smiech et al.,2021).

Evidence from Pakistan shows positive and negative shocks to oil prices can boost exports, especially in the textile industry, pointing to a complex relationship between energy price variations and export competitiveness. The price of oil has an impact on the cost of production and transportation, which in turn affects Sri Lanka's and other developing nations' ability to export. This increases the commodity prices, which influences to market prices. As Pakistan is a labour-abundant country, like Sri Lanka is affected by oil prices, though it does not depend much on machines (Kousar et al.,2022)

Yousefi and Wirjanto (2004) looked at how exchange rates affect the price of crude oil, with a particular emphasis on OPEC nations. Their empirical model, which employed the Generalized Method of Moments (GMM), showed how the exchange rate of the US dollar affects the pricing behavior of countries that export oil. Because Sri Lanka imports oil and has a fundamentally distinct economic structure, this study's applicability to the country is restricted, despite the fact that it employed strong econometric methodologies and offered insightful information. Oil price traumas vary by country. Higher oil prices are advantageous for oil-exporting countries, but they distress the economic performance of oil-importing countries. Nevertheless, international trade decreases the direct effects of oil price impacts on both types of countries. The results show that increasing oil prices harm all oil-importing countries, but trade has mitigated this effect in developed countries. Positive oil price shocks have a positive impact on developing oil-exporting countries, but trade has diminished this effect. However, the impact of the shocks on developed oil-exporting countries is minimal (Moshiri et al.,2024).

The cointegration link between oil and food prices in the examined countries (Olayungbo, 2021) reveals negative effects between food and oil prices, whereas positive effects exist over time. The findings imply that countries should pursue appropriate agricultural policies that promote favourable food prices and alternative energy options in order to ensure long-term food and oil supplies. A drop in oil prices has a negative impact on Azerbaijan's economy (Yildirim et al.,2021). There was a negative oil price shock that impacted the trade balance, causing currency depreciation and reducing economic activity. The 2014-16 oil price collapse substantially weakened the economic situation for oil-exporting countries. In Indonesia, (Lestari,2024) global crude oil prices had a small positive effect on the country's crude oil exports between 1992 and 2022, but this effect wasn't strong enough to be significant. Nonetheless, Crude oil prices are an external determinant that significantly affects Sri Lanka's trade performance.

2.4 Conclusion

Some studies applied the ARDL model (Ahmed et al., 2017), and some studies used the OLS method (Nadeem et al., 2012; Gurung and Rai, 2020; Lee et al., 2016), while some studies applied VECM (Bhavan, 2016; Khalid et al., 2021). Similar approaches confirm the effective use of the multiple linear regression model in macroeconomic export determinants with theoretical support. The studies Tharunya (2021) and George et al., (2022) emphasize how sudden global disruptions, such as COVID-19, cause sharp downturns in exports, affect the exchange rate, and impact macroeconomic stability, which is based on considering data from 2007 to 2024.

In conclusion, although existing literature provides foundational insights into how exchange rates, imports, and oil prices affect trade, it is often limited by two key shortcomings: the use of complex, less interpretable methodologies and the exclusion of recent and highly relevant data. The present study also considers crude oil price as an important factor affecting exports, since Sri Lanka heavily relies on importing crude oil, and when the crude oil prices increase, the production and transportation costs for exporters also increase. From 2019 to 2024, Sri Lanka experienced multiple overlapping crises, including the COVID-19 pandemic, foreign exchange shortages, a collapse in tourism earnings, widespread fuel and electricity shortages, inflation spikes, and a sovereign debt default in 2022. This study attempts to fill that gap by merging these variables into a comprehensive econometric framework. This study applies the MLR approach using monthly data from 2007 to 2024 to estimate the influence of selected macroeconomic variables on export performance in Sri Lanka. The findings are expected to contribute meaningfully to both academic discussions and real-world policy development in the country's export sector.

CHAPTER 3
Methodology

The main purpose of this study is to identify the impacts of imports, exchange rates, and crude oil prices on Sri Lanka's export performance from 2007 to 2024. This chapter presents the methodological framework employed, including the research design, data sources, and data preparation methods along with the statistical techniques applied to achieve the objectives of the study. Monthly time series data were collected and suitable transformations were used to ensure the consistency and validity. Multiple liner regression was applied as the main analytical technique, with diagnostic tests to verify the validity of the model assumptions.

3.1 Research Design

This study applies a quantitative research design to investigate the impacts of selected macroeconomic variables on Sri Lanka's exports. Secondary data collected from reliable national and international sources were used for this study. This research design provides a clear and organized approach to analyse the relationships between the selected independent variables and the export performance, providing evidence-based insights to guide policy decisions.

3.2 Data Collection

The study uses monthly time series data over the period January 2007 to December 2024. The dependent variable is exports, measured in US$ Million. Three predictor variables are employed: Imports (US$ Million), Exchange Rate (LKR/USD) and Crude Oil Price (US$ per barrel). Data on exports and imports were obtained from Central Bank of Sri Lanka (n.d). Exchange rate data were obtained from Federal Reserve Bank of St. Louis (n.d) and crude oil price data were collected from World Bank (n.d). A sample size of 216 was used in this study.

3.2.1 Variable description

- Exports: Sri Lanka's monthly total export value (US$ Million), covering all the sectors; agricultural exports, industrial exports, mineral exports and other unclassified exports.

- Imports: Sri Lanka's monthly total import value (US$ Million), covering consumer goods, intermediate goods and investment goods and other unclassified imports.

- Exchange Rate: Nominal exchange rate: Sri Lankan rupee to one US dollar (spot rate).

- Crude Oil Price: Monthly average crude oil price (US$ per barrel).

3.3 Methodology

3.3.1 Preliminary Analysis

As the first step, we get descriptive statistics of all variables to get a thorough understanding of the dataset. This includes measures of central tendency (mean, median), dispersion (standard deviation, range) and distributional properties (skewness, kurtosis). Next, we generate time series plots for each variable to investigate the trends and seasonal fluctuations in the data. Finally, we compute correlation matrix of all variables to evaluate the pairwise correlation of independent variables with exports and identify which variable affects exports and to what extent.

3.3.2 Stationarity Testing

In time series regression, it is required that all the variables used in the model are stationary to generate valid results. Therefore, we use Augmented Dickey- Fuller (ADF) test to check the stationarity of each variable.

We apply the ADF test to each of the four variables-exports, imports, exchange rate and crude oil price in the following forms:

1. The original series is used to check whether all variables are already stationary without needing any transformation.
2. Log transformed series is used because a log transformation can stabilize variance and reduce the influence of large fluctuations and in return can improve the stationarity of data.
3. First-differenced series is used because differencing removes trend component and improves stationarity.
4. First differences of log-transformed series is used because this removes trend component, stabilizes variance and allows the variables to be interpreted in growth rate terms.

Hypothesis of ADF test:

- Null hypothesis (**H_0**): The variable has a unit root, meaning it is non-stationary.
- Alternative hypothesis (**H_1**): The variable does not have a unit root, meaning it is stationary.

Decision rule:

If the p-value < 0.05 the null hypothesis of a unit root is rejected, indicating that the series is stationary.

For the model, we choose the first differences of log transformed series which ensures the stationarity of all variables and allows the interpretation of results in terms of growth rates (short-run percentage changes).

3.4 Model Specification

Based on the transformed stationary variables the multiple linear regression model is specified as,

$$\Delta \ln(Exports_t) = \beta_0 + \beta_1 \Delta \ln(Imports_t) + \beta_2 \Delta \ln(ExchangeRate_t) + \beta_3 \Delta \ln(CrudeOilPrice_t) + \varepsilon_t$$

In this model:

Δln denotes the first difference of the natural logarithm of each variable.

Δ ln(Exports): Growth rate of Exports at time t.
Δ ln(Imports$_t$): Growth rate of Imports at time t.
Δ ln(ExchangeRate$_t$): Growth rate of Exchange Rate at time t.
Δ ln(CrudeOilPrice$_t$): Growth rate of Crude Oil Price at time t.
β_0: Intercept
β_1, β_2, β_3: Estimated coefficients
ε_t: Error term

This model allows us to interpret the coefficients as short-run elasticities representing the percentage change in exports associated with a 1% change in each predicter variable.

Coefficient interpretations:

β_0: Average export growth rate when all predictors are not changing (Δln=0).
β_1, β_2, β_3: short run elasticities (percentage change in exports associated with a 1% change in respective predictor variable holding other variables constant).

Hypotheses regarding coefficients:

Hypothesis 1:

- **H_0:** $\beta_1 = 0$ (no short run relationship between the growth rate of Imports and growth rate of Exports).
- **H_1**: $\beta_1 \neq 0$ (there is a statistically significant relationship between the growth rate of Imports and growth rate of Exports).

Hypothesis 2:

- **H_0:** $\beta_2 = 0$ (no short run relationship between the growth rate of Imports and growth rate of Exports).

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- **H_1:** $\beta_2 \neq 0$ (there is a statistically significant relationship between the growth rate of Exchange Rate and growth rate of Exports).

Hypothesis 3:

- **H_0:** $\beta_3 = 0$ (no short run relationship between the growth rate of Imports and growth rate of Exports).
- **H_1**: $\beta_3 \neq 0$ (there is a statistically significant relationship between the growth rate of Crude Oil Price and growth rate of Exports).

3.5 Model Diagnostics

After the estimation, we evaluate the model using key diagnostic tests on residuals.

3.5.1 Autocorrelation Testing:

Breusch-Godfrey (BG) LM test is used to check whether the residuals are serially correlated. This test is chosen because we consider monthly data.

The hypotheses of the test are as follows:

- **H_0**: No serial correlation up to the chosen lag order.
- **H_1:** Serial correlation exists in the residuals up to the chosen lag order.

Decision rule is as follows:
If p-value > 0.05, we fail to reject null hypothesis indicating that there is no evidence of autocorrelation.

We test the autocorrelation at lag 1 and lag 12. Lag 1 is used to check the short run autocorrelation. Lag 2 is used to check the seasonal autocorrelation.

3.5.2 Heteroskedasticity Testing:

We perform studentized Breusch–Pagan test (approximating White's test using squared and interaction terms) to check whether the variance is constant over time. Studentized Breusch–Pagan test is used because we found evidence of autocorrelation in residuals.

The hypotheses of the test are as follows:

- **H_0**: The variance of the residuals is constant (homoscedastic).
- **H_1**: The variance of the residuals is heteroskedastic, i.e., it depends on one or more predictors or their functions (squares, interactions, etc.).

Decision rule is as follows:
If p-value < 0.05 we reject the null hypothesis confirming the evidence of heteroskedasticity.

3.5.3 Multicollinearity Testing:

We use Variance Inflation Factor (VIF) to check whether the explanatory variables are highly correlated with each other.

Interpretation of VIF value is as follows:
VIF value approximately equal to 1 suggest there is no multicollinearity and VIF value > 5 (or 10) suggests possible multicollinearity.

3.5.4 Normality Testing of Residuals:

We use Jarque–Bera test and Q-Q plot of residuals to check the normality assumption of residuals.

The hypotheses of the test are as follows:

- **H_0**: Residuals are normally distributed.
- **H_1**: Residuals are not normally distributed.

Decision rule is as follows:

If p- value < 0.05, we reject the null hypothesis indicating that the residuals are not normally distributed.

The above four diagnostics evaluate the key assumptions of the Ordinary Least Square (OLS) regression for time series data.

CHAPTER 4
Data Analysis

4.1 Preliminary analysis

Table 1: Descriptive statistics of variables

Variable	Mean	Median	SD	Minimum	Maximum	Range	Skewness	Kurtosis
Exports	878.24	898.24	172.96	282.31	1248.77	966.46	-0.48	-0.07
Imports	1471.42	1523.49	313.71	606.31	2240.98	1634.67	-0.54	-0.20
Exchange Rate	169.56	143.44	72.61	107.61	363.95	256.34	1.46	0.91
Crude Oil Price	75.63	74.35	23.23	21.04	132.83	111.79	0.09	-0.84

Note: This table presents the selected descriptive statistics, including mean, median, standard deviation (SD), minimum, maximum, range, skewness, and kurtosis for the dependent variable and the independent variables.

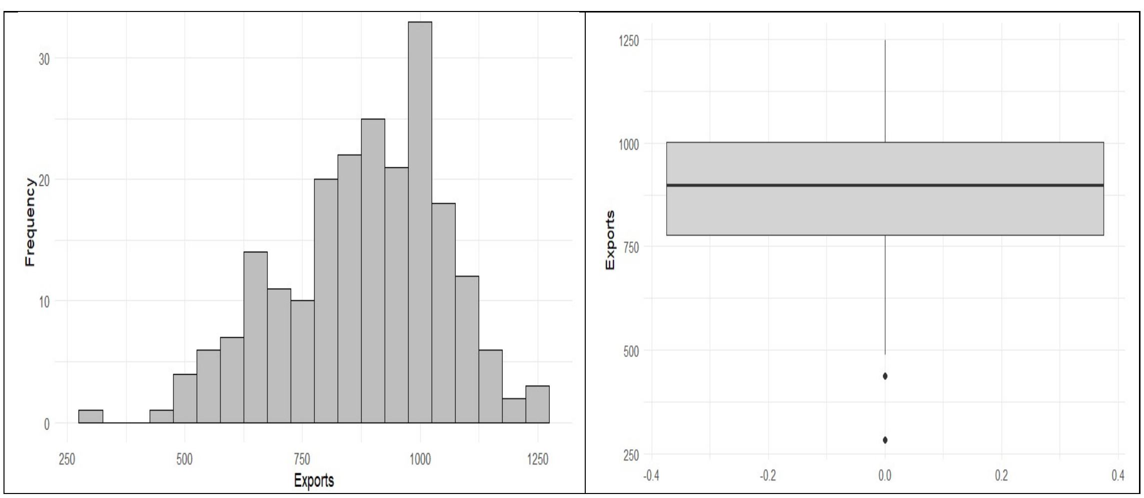

Figure 1: Histogram (left) and boxplot (right) for Exports

Notes: This figure shows the histogram and the boxplot for the monthly values of exports from 2007 to 2024.

The average monthly value of exports during the period 2007 to 2024 is USD 878.24 million, with a median of USD 898.24 million. This indicates that the distribution of monthly exports is slightly negatively skewed (skewness = -0.48). This can be observed in both the histogram and the boxplot (Figure 1). The standard deviation of exports is USD 172.96 million. This indicates the monthly export values vary by approximately 173 units from the average. The distribution of monthly exports is approximately mesokurtic (kurtosis = -0.07), implying a moderate peak with a range of USD 966.46 million.

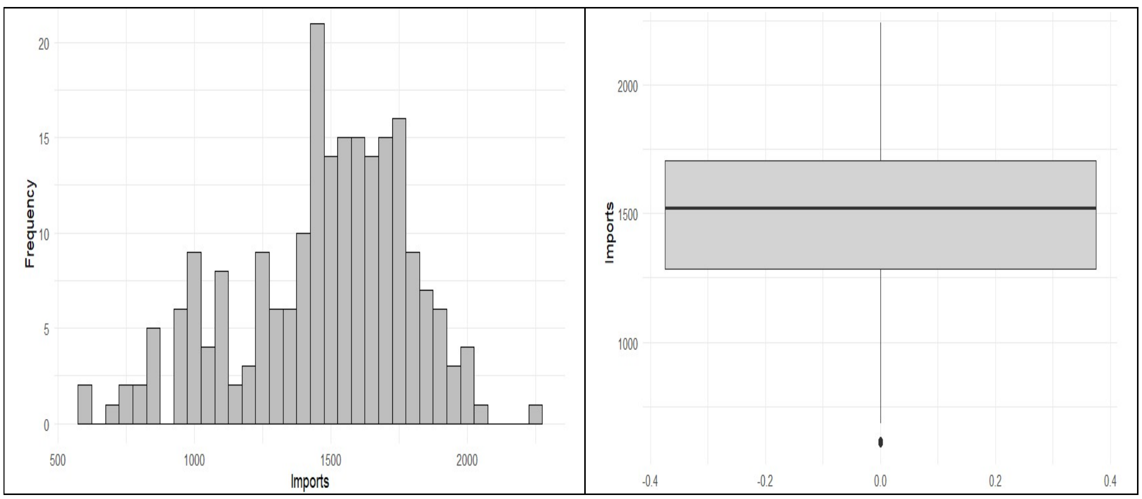

Figure 2:Histogram (left) and boxplot (right) for Imports

Notes: This figure shows the histogram and the boxplot for the monthly values of imports from 2007 to 2024.

Monthly imports have the highest average among all the other variables (mean = USD 1471.42 million) with a median of USD 1523.49 million. This implies that the distribution of monthly imports is negatively skewed (skewness = -0.54). Figure 2 confirms this. Moreover, monthly imports have the largest standard deviation (SD = USD 313.71 million) among all the other variables. This means monthly import values vary by approximately 314 units from the average. Monthly imports have an approximately platykurtic distribution (kurtosis = -0.20) with a range of USD 1634.67 million.

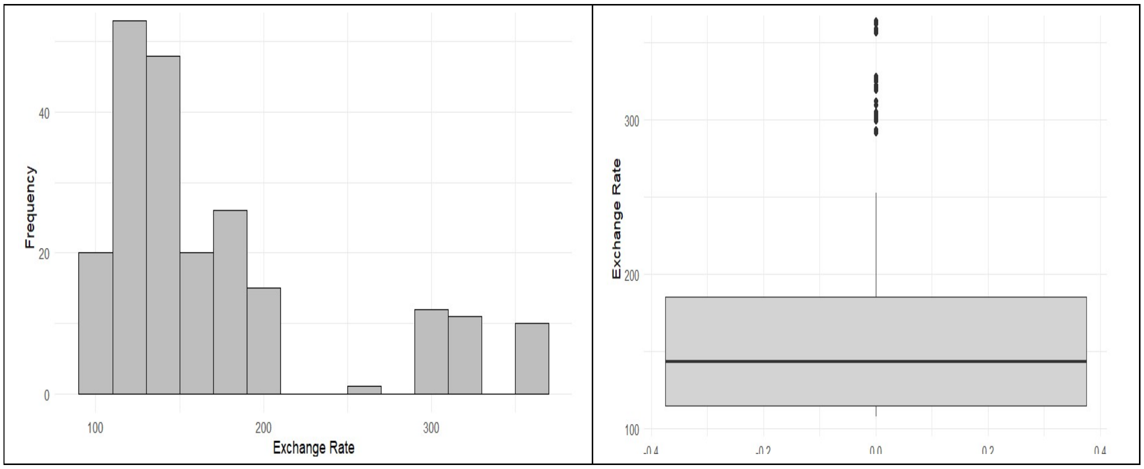

Figure 3: Histogram (left) and boxplot (right) of Exchange Rate

Notes: This figure shows the histogram and the boxplot for the monthly values of exchange rate from 2007 to 2024.

The average monthly value of exchange rate during the study period is USD 169.56 million, with a median of USD 143.44 million. The standard deviation of exports is USD 72.61 million. This indicates the monthly export values vary by approximately 73 units from the average. The distribution of monthly exports is approximately leptokurtic (kurtosis = 0.91), implying a slightly more peaked distribution with a range of USD 256.34 million. The histogram in Figure 3 shows that there are two separate groups of monthly exchange rate values.

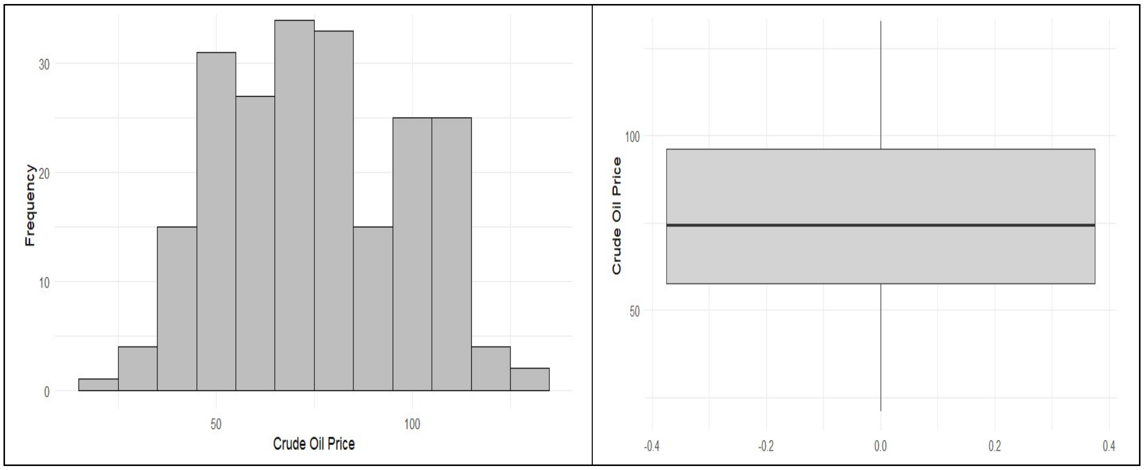

Figure 4: Histogram (left) and boxplot (right) of Crude Oil Price

Notes: This figure shows the histogram and the boxplot for the monthly values of crude oil price from 2007 to 2024.

Monthly crude oil prices have the lowest average among all the other variables (mean = USD 75.63 million) with a median of USD 74.35 million, indicating that the distribution of monthly crude oil prices is nearly symmetric (skewness = 0.09). This can be observed in both the histogram and the boxplot (Figure 4). Furthermore, among the all the other variables, the lowest standard deviation can be seen in this variable (SD = USD 23.23 million). Monthly crude oil prices have a platykurtic distribution (kurtosis = -0.84) with a range of USD 111.79 million.

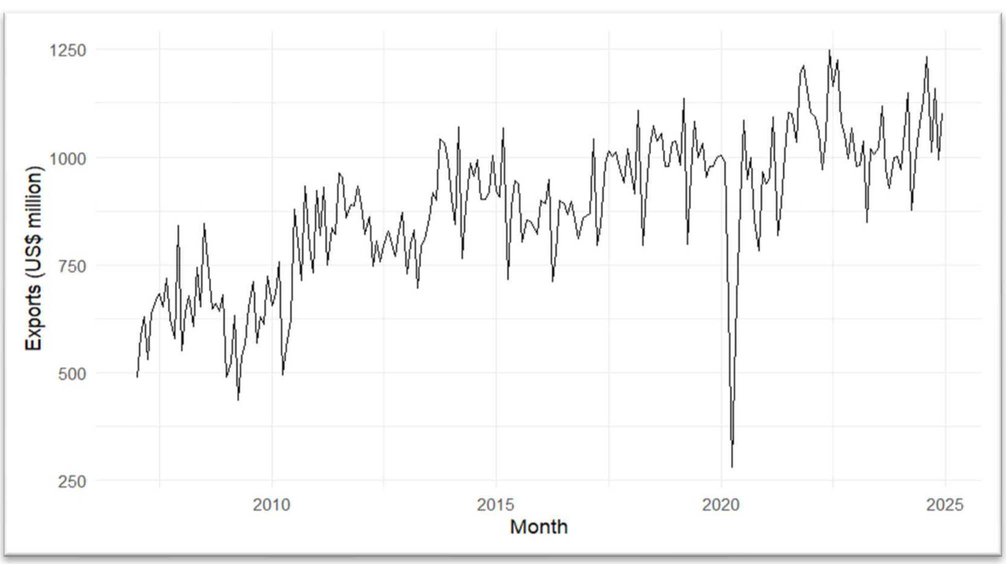

Figure 5: Time series plot of Monthly Export Performance (2007-2024)

Notes: This figure shows the time series plot for the monthly export performance over the period 2007 to 2024.

In general, Sri Lanka's monthly exports show an upward trend over time although the growth is fluctuating and limited. Exports have experienced a sharp decline around 2020, which is probably due to the pandemic related disruptions. In recent years exports have fluctuated within the range of US$ 1000-1200 million.

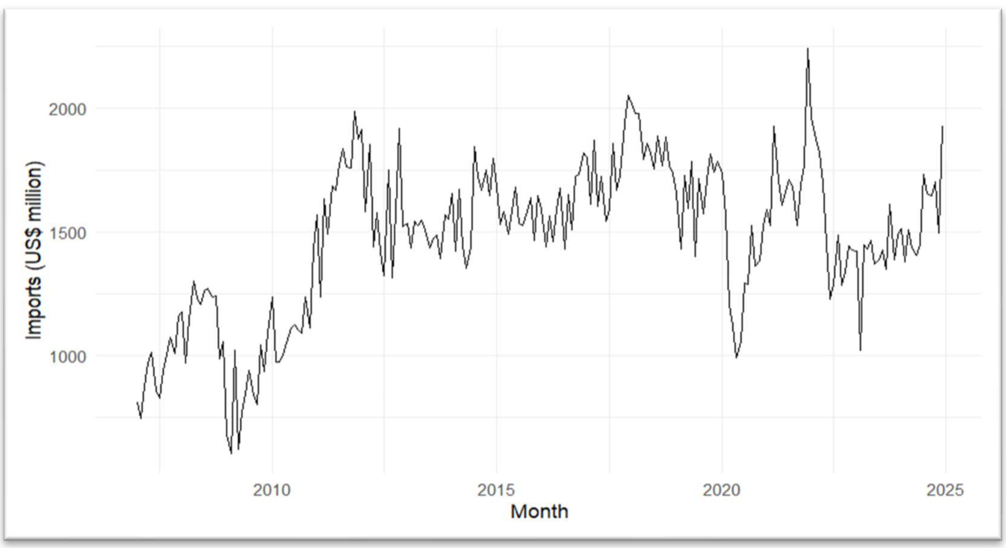

Figure 6: Time series plot of Monthly Import Performance (2007-2024)

Notes: This figure shows the time series plot for the monthly import performance over the period 2007 to 2024.

We can see that imports remained higher than exports throughout the whole period. This indicates a trade deficit. Imports have increased steadily up to around 2011- 2012 and after that they have fluctuated between US$ 1500-2000 million. A significant drop in imports can be seen around 2020 and 2022 though a partial recovery followed.

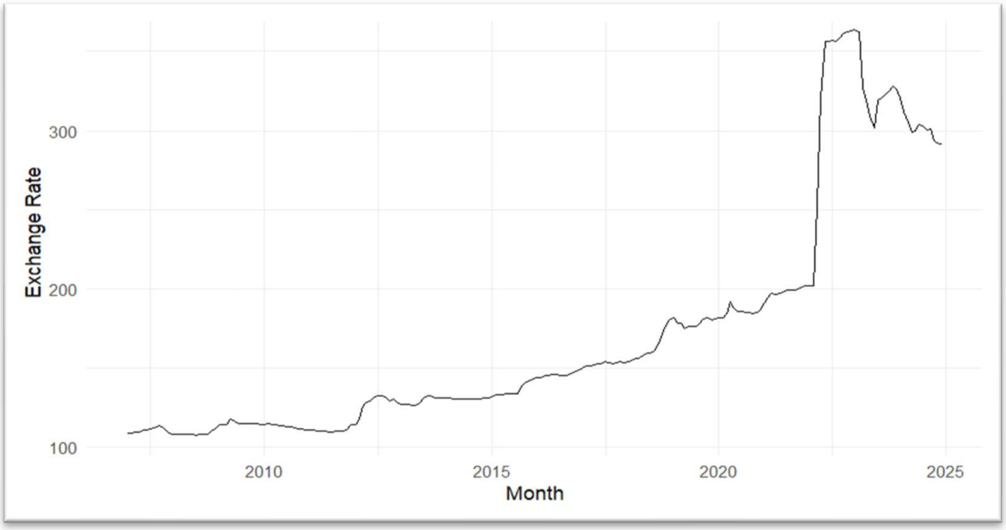

Figure 7: Time series plot of Monthly Exchange Rate (2007-2024)

Notes: This figure shows the time series plot for the values of monthly exchange rate over the period 2007 to 2024.

We can see that from 2007 to around 2015, the currency remained relatively stable around 100-150. This was followed by a gradual depreciation and a sharp rise in 2022. Although it has reduced somewhat since then it continues to remain at high values compared to earlier years.

Figure 8: Time series plot of Monthly Crude Oil Price (2007-2024)

Notes: This figure shows the time series plot for the values of monthly crude oil price over the period 2007 to 2024.

Crude oil prices were highly volatile, experiencing a sharp increase around 2008 before dropping sharply. Then they showed a fluctuating upward trend up until experiencing another sharp decline around 2015. We can see another sharp decrease in 2020 which can be attributed to pandemic and then again, a sharp rise in 2022 before gradually decreasing.

Correlation Matrix

Table 2: Correlation matrix of the study variables

Variable	Exports	Imports	Exchange Rate	Crude Oil Price
Exports	1.00	0.68	0.58	0.09
Imports	0.68	1.00	0.16	0.07
Exchange Rate	0.58	0.16	1.00	0.01
Crude Oil Price	0.09	0.07	0.01	1.00

Table 2 presents the correlation matrix of the study variables using the monthly data over the period 2007 to 2024. According to the results, exports are positively correlated with imports (r = 0.68), suggesting that when the exports increase, imports also increase. Exports are positively correlated with the exchange rate (r = 0.58), which means that when exports increase, the exchange rate tend to increase as well. Furthermore, crude oil prices show very weak correlations with exports (r = 0.09), imports (r = 0.07), and the exchange rate (r = 0.01). According to the results of the above table, imports show a weak positive correlation with the exchange rate (r = 0.16). Based on these results, the strongest relationships are observed between exports, imports, and the exchange rate, while the crude oil prices are largely uncorrelated with the other study variables.

Table 3: Correlation matrix of log-transformed variables

Variable	$\ln(Exports_t)$	$\ln(Imports_t)$	$\ln(ExchangeRate_t)$	$\ln(CrudeOilPrice_t)$
$\ln(Exports_t)$	1.00	0.72	0.58	0.16
ln(Imports	0.72	1.00	0.28	0.10
$\ln(ExchangeRate_t)$	0.58	0.28	1.00	-0.04
$\ln(CrudeOilPrice_t)$	0.16	0.10	-0.04	1.00

Since the distributions of some variables were skewed, a log transformation was applied, and the correlation matrix was re-estimated (Table 3). According to these results, log-transformed exports are positively and strongly correlated with log-transformed imports (r = 0.72). Log-transformed exports show a moderate positive correlation with the exchange rate (r = 0.58), and a weak but positive correlation with the crude oil prices (r = 0.16). Furthermore, log-transformed imports also show a weak positive correlation with both the log-transformed exchange rate (r = 0.28) and the log-transformed crude oil prices (r = 0.10). However, the log-transformed of exchange rates show a negative correlation with the log-transformed of crude oil prices (r = -0.04).

Based on these results, the correlation matrix after log transformation remains generally similar to that of the original variables, with exports show the strongest correlation with imports and the exchange rate.

4.2 Stationarity Testing

Table 4: Unit root test results

Variable	Dickey-Fuller Statistic	Lag Order	p-value	Stationarity (α=0.05)
Exports	-3.78	5	0.02	Yes
Imports	-3.27	5	-3.27	No
Exchange Rate	-2.45	5	0.39	No
CrudeOil Price	-2.91	5	0.19	No

Notes: This table presents the results of the Augmented Dickey-Fuller (ADF) unit root test conducted for exports, imports, exchange rate, and crude oil price. The test was applied at a 5% level of significance (α=0.05) with a lag order of 5.

According to the Table 4, only exports are stationary at their levels, as the ADF statistic (-3.78) is significant with a p-value of 0.02, under the 5% level. Moreover, imports, exchange rate, and crude oil price are non-stationary, with p-values of -3.27, 0.39, and 0.19 respectively.

Since only exports are stationary, differencing or transformation are needed to the variables before further analysis.

Table 5: Unit root test results for log-transformed variables

Variable	Dickey-Fuller Statistic	Lag Order	p-value	Stationarity (α=0.05)
$\ln(Exports_t)$	3.93	5	0.01	Yes
ln(Imports)	3.23	5	0.08	No
$\ln(ExchangeRate_t)$	2.50	5	0.37	No
$\ln(CrudeOilPrice_t)$	2.80	5	0.24	No

Notes: This table presents the unit root test results for the log-transformed variables using the ADF test with a lag order of 5 at a 5% significance level.

Based on the results of the Table 5, only log-transformed exports are stationary, with an ADF statistic of 3.93 and a p-value of 0.01. Log-transformed imports, log-transformed exchange rate, and log-transformed crude oil price are non-stationary, with p-values 0.08, 0.37, and 0.24 respectively.

As the log transformation alone does not make all the variables stationary, getting the first difference of the variables is needed before further analysis.

Table 6: Unit root test results for the first difference of variables

Variable	Dickey-Fuller Statistic	Lag Order	P-value	Stationary (α=0.05)
$\Delta\,(Exports_t)$	-10.10	5	<0.01	Yes
$\Delta\,(Imports_t)$	-7.33	5	<0.01	Yes
$\Delta\,(ExchangeRate_t)$	-5.46	5	<0.01	Yes
$\Delta\,(CrudeOilPrice_t)$	-6.64	5	<0.01	Yes

Notes: This table shows the unit root test results for the first-differenced series using the ADF test with a lag order of 5 at a 5% level of significance.

Table 6 shows that, all the study variables become stationary after taking the first differences, with p-values below 0.01 and with the ADF statistics -10.10, -7.33, -5.46, and -6.64 respectively.

Table 7: Unit root test results for the first difference of log-transformed variables

Variable	Dickey-Fuller Statistic	Lag Order	P-value	Stationary (α=0.05)
Δ ln($Exports_t$)	-10.20	5	<0.01	Yes
Δ ln($Imports_t$)	-7.33	5	<0.01	Yes
Δ ln($ExchangeRate_t$)	-5.46	5	<0.01	Yes
Δ ln($CrudeOilPrice_t$)	-6.36	5	<0.01	Yes

Notes: This table presents the unit root test results for the first-differenced log-transformed series using the ADF test with a lag order of 5 at a 5% level of significance.

According to the Table 7, log-transformed exports, log-transformed imports, log-transformed exchange rate, and log-transformed crude oil price are stationary after first differencing, with p-values below 0.01 and with the ADF statistics -10.20, -7.33, -5.46, and -6.36 respectively.

4.3 Model diagnostics

After the estimation of the model,

$\Delta \ln(Exports_t) = \beta_0 + \beta_1 \Delta \ln(Imports_t) + \beta_2 \Delta \ln(ExchangeRate_t) + \beta_3 \Delta \ln(CrudeOilPrice_t) + \varepsilon_t$

model diagnostics were performed to check for violation of key assumptions and to identify whether re-estimation of the model is required.

a) **Autocorrelation Testing**:

Breusch-Godfrey test for serial correlation of order up to 12

LM test = 93.789, df = 12, p-value = 9.083e-15

Decision: Since p-value < 0.05, we reject the null hypothesis of no serial correlation at 5% level
significance.

Conclusion: There is serial correlation up to the 12^{th} lag order.

Breusch-Godfrey test for serial correlation of order up to 1

LM test = 22.924, df = 1, p-value = 1.686e-06

Decision: Since p-value < 0.05, we reject the null hypothesis of no serial correlation at 5% level significance.

Conclusion: There is serial correlation up to the first lag order.

b) **Heteroskedasticity Testing**:

studentized Breusch-Pagan test
LM test = 22.924, df = 1, p-value = 1.686e-06

Decision: Since p-value < 0.05, we reject the null hypothesis of constant variance at 5% level significance.

Conclusion: There is evidence of heteroskedasticity.

c) **Multicollinearity Testing:** VIF

$\Delta\ln(Imports_t)$	$\Delta\ln(ExchangeRate_t)$	$\Delta\ln(CrudeOilPrice_t)$
1.031290	1.017870	1.013565

All the VIF values are approximately equal to 1.

Conclusion: There is no multicollinearity. The predictor variables are not correlated with each other.

d) **Normality Testing of Residuals**:

Jarque Bera Test
X-squared = 92.016, df = 2, p-value < 2.2e-16

Decision: Since the p-value < 0.05, we reject the null hypothesis of normality of residu als at 5% level of significance.

Conclusion: The residuals are not normally distributed.

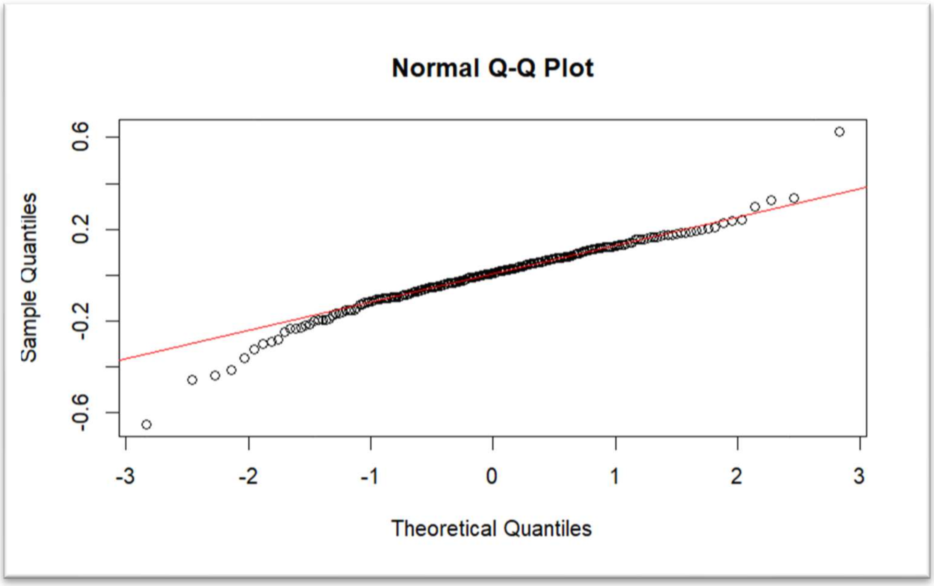

Figure 9: Q-Q plot of residuals

In the middle part, most points lie closer to the line of theoretical normal distribution, suggesting that the residuals are approximately normal in the centre of the distribution.

Points deviate from the line at lower tail (left side) and upper tail (right side). This indicates minor deviations from normality.

We can see several points at the tail deviate significantly, indicating potential outliers or slight non normality in residuals.

4.4 Model summary

4.4.1 Summary 1 (not addressed the autocorrelation and heteroskedasticity issues)

Residuals:

Min	Q1	Median	Q3	Max
-0.65320	-0.07799	0.00764	0.08931	0.62719

Coefficients:

	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	0.003255	0.010233	0.318	0.75072

Δ ln($Imports_t$)	0.341733	0.078870	4.333	2.28e-05 ***
Δ ln($ExchangeRate_t$)	-0.295761	0.377630	-0.783	0.43439
Δ ln($CrudeOilPrice_t$)	0.372210	0.101735	3.659	0.00032 ***

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.1477 on 211 degrees of freedom (1 observation deleted due to missingness)

Multiple R-squared: 0.1537, Adjusted R-squared: 0.1417

F-statistic: 12.78 on 3 and 211 DF, p-value: 1.051e-07

Since we observe violations of no autocorrelation assumption and homoskedasticity assumption we re-estimate the model using Newey-West standard errors which address these issues.

But the re-estimation of the model does not change the residuals, coefficients or R^2(model fit). It only changes the standard errors and inference regarding coefficient significance.

Residuals interpretation:

The range of residuals is relatively small (range = 1.28039). Median is approximately equal to zero. Q1 and Q3 values are relatively close in magnitude, indicating that the residuals are approximately symmetric. Values of Q1, median and Q2 suggest that there is no extreme skewness. The min is slightly more extreme than max, indicating a mild left skewness in the distribution.

Conclusion: In general, the residuals are nearly symmetric, but on the left side we can see a mild skewness. The Q-Q plot confirms the same.

Normality assumption of residuals:

Although Jarque–Bera test indicates a violation of normality assumption, residuals and Q-Q plot suggests only mild deviations from normality. Besides, in large samples Central Limit Theorem (CLT) ensures that coefficient estimates are approximately normally distributed even if residuals are not perfectly normal. Normality is mainly required for small sample exact inference.

Therefore, the re-estimated model's coefficients are valid considering autocorrelation issues, heteroskedasticity issues as well as normality assumption of residuals.

4.4.2 Summary 2 (Model is re-estimated: autocorrelation and heteroskedasticity issues are addressed using Newey-West standard errors)

t test of coefficients:

	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	0.0032553	0.0085871	0.3791	0.7050
Δ ln($Imports_t$)	0.3417328	0.0818924	4.1730	4.395e-05 ***
Δ ln($ExchangeRate_t$)	-0.2957610	0.2842149	-1.0406	0.2992
Δ ln($CrudeOilPrice_t$)	0.3722098	0.2459323	1.5135	0.1317

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$$\Delta \ln(Exports_t) = \beta_0 + \beta_1 \Delta \ln(Imports_t) + \beta_2 \Delta \ln(ExchangeRate_t) + \beta_3 \Delta \ln(CrudeOilPrice_t) + \varepsilon_t$$

Interpretations of coefficients:

β_0 (Intercept): 0.0032553 ($p = 0.7050$)

When all independent variables' changes are zero average short run export growth rate is about 0.33%. But this is not statistically significant.

β_1 (coefficient of Δ ln($Imports_t$)): 0.3417328 ($p = 4.395e-05$)

When import growth increases by 1%, there is a 0.34% increase in export growth in the short run, holding other variables constant. Since p-value < 0.01 we reject, H_0: $\beta_1 = 0$. This relationship is statistically significant.

β_2 (coefficient of Δ ln(ExchangeRate$_t$)): -0.2957610 (p = 0.2992)

When there is a 1% depreciation (positive growth in Exchange Rate), Export growth decreases by 0.30% in short run, holding other variables constant. Since p-value > 0.05 we fail to reject, H_0: $\beta_2 = 0$. This relationship is not statistically significant.

β_3 (coefficient of Δln(CrudeOilPrice$_t$)): 0.3722098 (p = 0.1317)

When there is 1% increase in Crude Oil Price, Export growth increases by 0.37% in the short run, holding other variables constant. Since p-value > 0.05 we fail to reject, H_0: $\beta_3 = 0$. This relationship is not statistically significant.

4.4.3 Model Fit

Adjusted R squared = 0.1417

This implies that the model explains about 14.17% of the variation in export growth.

p-value = 1.051e-07

The overall model is statistically significant. That is, at least one predictor affects exports.

4.5 Summary of interpretations

- Import growth positively affects Export growth in short run, and this relationship is statistically significant.
- Exchange rate growth (domestic currency depreciation) has a negative but statistically insignificant relationship with export growth in the short-run.
- Crude Oil Price growth positively affects Export growth in short run, but this relationship is not statistically significant.
- While the overall model is statistically significant, the model's explanatory power is limited. This indicates that there are unobserved factors that influence export performance.

CHAPTER 5
DISCUSSION

This study used monthly data from 2007 to 2024 to examine the short-term factors influencing Sri Lanka's exports. According to the data, the exchange rate has a negative but statistically negligible impact on export growth, while imports have a positive and statistically significant influence. Crude oil prices also show a positive but statistically insignificant influence. Although the model is overall statistically significant, it explains only about 15 percent of the variation in export growth, suggesting that other unobserved factors also influence export performance.

The coefficient for $\Delta \ln (Imports_t)$ is 0.341733, with a p-value of 4.395e-05. Since this p-value is less than 0.01, it is statistically significant. This means a 1% increase in import growth leads to a 0.34% increase in export growth, holding other variables constant. This relationship is positive and significant. The positive relationship between imports and exports highlights the dependence of Sri Lanka's export sector on imported inputs such as intermediate and investment goods. The conclusion that imports expansion frequently enhances export growth in developing economies rather than restricts it is supported by this observation.

The literature has identified various ways through which imports can affect the export of the country. Importing new and advanced goods can reduce production constraints, increasing firm productivity (Halpern *et al.*, 2005). Higher productivity firms may encounter export costs or develop new products for foreign customers (Bas and Strauss-Khan, 2011). According to the firm-level evidence available for Italy, if learning by exporting takes place and imports have a direct and positive impact on the likelihood of exporting, the entire nation may experience significant feedback effects because of the productivity gains that have been documented as a result of the firm's export activity.

The previous findings indicate that increased imports of final goods increase market competition, encouraging import-substituting firms to engage in innovative activities. The advanced countries' technologies embodied in imports are transferred to an importing country, increasing its productivity regardless of the relevant product categories (Uddin and Khanam,2017). Exports of agricultural products are also dependent on input supply because when there are more inputs available at a lower cost and efficient technology to use them to produce that much commodity or good that can be exported, agriculture productivity may increase (Gilani,2015). Japan accounts for 5.7% of global imports. Other countries account for less than 4% of imports. Imports reflect both resource availability and price demand. Countries usually import such products that are difficult to produce due to a lack of resources or their high cost. Products should be available for export in order to balance imports (Madzinova,2011).

The coefficient for Δ ln ($ExchangeRate_t$) is -0.295761, with a p-value of 0.2992. Since this p-value is greater than 0.05, the relationship is not statistically significant. This means there is no significant relationship between the exchange rate and export growth in this model. The exchange rate had a negligible but adverse effect on exports. Domestic currency depreciation usually increases export competitiveness by lowering the price of domestic goods in overseas markets. However, in Sri Lanka's case, depreciation may raise the cost of imported inputs, thereby reducing the overall benefits for exporters. Furthermore, structural rigidities, high reliance on imported raw materials, and exchange rate pass-through effects could be the reasons why currency fluctuations haven't resulted in appreciable export increases.

Examine the effects of exchange rate uncertainty on exports, as well as how accounting for exchange rate uncertainty influences export response to exchange rate shocks. Exchange rate uncertainty has a negative and significant impact on US exports. Exchange rate uncertainty tends to strengthen exports' dynamic response to exchange rate shocks, and exports respond asymmetrically to both positive and negative exchange rate shocks of equal magnitude (Rahman and Serletis,2009).

The exchange rate volatility of exports to China only affects plastic goods, although many goods experience negative effects due to exchange rate depreciation. In India, exchange rate volatility affects the largest number of export commodities (Sugiharti *et al*.,2020). Rahman and Serletis, (2009) reported that exchange rate uncertainty has a significant statistical and economic impact on exports. The exchange rate uncertainty tends to amplify exports' negative dynamic response to a positive exchange rate shock.

The coefficient for Δ ln($CrudeOilPrice_t$) is 0.372210, with a p-value of 0.1317. This p-value is greater than 0.05, making it statistically not significant. This indicates that a 1% increase in crude oil price leads to increase in export growth 0.37% in short run, holding other variables constant. There are two potential explanations for this outcome. First, higher oil prices may lead to increased demand for Sri Lankan exports such as tea, rubber, and garments, especially in oil-exporting countries that experience greater liquidity. Second, trade patterns are frequently influenced by global commodity cycles, and increased oil prices may indirectly increase demand for Sri Lankan commodities and foreign exchange inflows.

Crude oil plays a major role in affecting U.S. exports, mainly because restrictions on crude shipments limit the ability to sell directly to international markets. As a result, exporting refined petroleum products like gasoline, diesel and petrol has become increasingly important for the US. Additionally, these limits maintain U.S. crude prices below global averages, which lowers producer earnings and negatively affects the trade balance. Cheaper oil may help refiners, but it costs producers money and investment possibilities. If export restrictions were eased, U.S. crude oil could enter global markets more freely, helping producers gain higher prices and expand production. This would boost U.S. competitiveness and geopolitical influence in the global energy sector in addition to improving the country's trade balance. Therefore, the United States has immediate economic and strategic effects from decisions on crude oil policy (Brown *et al*., 2014).

Herath *et al*., (2013) reported that Sri Lanka's exports are asymmetrically impacted by crude oil prices. The results show that whereas positive oil price shocks primarily impact interest rates and foreign reserves, they have no apparent effect on exports due to government protections such price controls, subsidies, and state-to-state agreements on oil imports. However, because lower oil prices cut production and transportation costs, making Sri Lankan goods more competitive internationally, negative oil price shocks have a bigger and more positive impact on exports. The impulse response analysis further indicates that the maximum positive impact of a crude oil price decrease on exports occurs about seven to eight months after the shock. Although policy buffers mainly absorb price rises, drops in oil prices obviously improve Sri Lanka's export performance.

Amarasinghe *et al*., (2018) assert that changes in the price of crude oil have a substantial impact on the Sri Lankan economy, which in turn affects the export performance of the country. Since Sri Lanka is highly dependent on imported petroleum, rising oil prices increase production and transportation costs, leading to higher inflation through cost-push effects. Export-oriented industries are immediately impacted by this circumstance as it raises their cost structures and lowers their ability to remain competitive on the global market. The study also found that there is a substantial long-term negative correlation between crude oil prices and inflation, as determined by the Colombo Consumer Price Index. In the short term, shocks to oil prices take almost three months to affect the economy and almost eight months to stabilise. Based on these dynamics, it appears that Sri Lanka's exporters face uncertainty due to fluctuations in the world's crude oil markets, which impedes steady export development.

Since imports significantly drive exports, restrictive import policies may harm export performance. Similarly, authorities shouldn't rely on currency depreciation alone to promote exports, given the exchange rate's negligible role. Furthermore, the statistically insignificant impact of crude oil prices highlights the need for policies that strengthen resilience against global energy price volatility. Therefore, greater emphasis should be placed on export base diversification, technological advancement, and structural reforms.

One unexpected result was the lack of a significant relationship between exchange rate movements and exports. Depreciation should encourage exports, according to conventional trade theory, but Sri Lanka's reliance on imported inputs appears to counteract this effect. This highlights how crucial it is to consider imports and exports as interconnected elements of the manufacturing process rather than as distinct phenomena.

Overall, the findings imply that supply-side variables and international circumstances have a greater influence on Sri Lanka's export development than currency manipulation. The findings reinforce the need for long-term strategies that strengthen competitiveness, expand industrial capacity, and reduce vulnerability to external shocks. To provide a more thorough explanation of export success, future studies should expand the analysis by including new variables such foreign direct investment, global demand, and changes in domestic policy.

CHAPTER 6
Conclusion

6.1 Summary

This study contributes to the existing literature on Sri Lanka's export performance and short-run macroeconomic dynamics in several ways. First, it provides updated empirical evidence covering the period 2007–2024, capturing the effects of recent economic shocks, policy changes, and global volatility. Second, by examining the combined roles of exchange rates, imports, and crude oil prices, the study adds a more integrated perspective compared to earlier research that often considered these factors in isolation. Third, the use of time series econometric techniques on first-differenced log-transformed variables offers robust short-run insights, complementing studies that focus primarily on long-run relationships. Overall, the findings deepen our understanding of how external shocks and domestic trade dependencies shape export performance in a small open economy like Sri Lanka. They provide a bridge between theoretical models of export dynamics and practical, policy-relevant insights, offering a basis for future research on trade and macroeconomic policy in similar developing economies.

6.2 Main Findings

This study analysed the short-term factors influencing Sri Lanka's exports using monthly data from 2007–2024. This study set out to answer four key questions about Sri Lanka's export performance. First, it asked how imports influence exports. The results show a strong and positive relationship: when imports increase by 1%, exports rise by about 0.34% in the short run. This reflects the heavy reliance of Sri Lanka's export industries on imported raw materials, machinery, and intermediate goods, making imports a crucial driver of export activity.

The second question focused on the effect of exchange rate fluctuations. The analysis found that while a depreciation of the Sri Lankan rupee appears to reduce exports slightly (around 0.30%), this effect is statistically insignificant. In other words, exchange rate movements alone do not explain changes in export performance, suggesting that structural constraints and competitiveness issues limit the benefits of currency depreciation.

The third question examined how global crude oil prices affect exports. The findings show a statistically insignificant positive effect: a 1% increase in crude oil prices leads to about a 0.37% rise in export growth.

Finally, the study considered which of these factors exerts the most significant influence on Sri Lanka's exports. The results reveal that imports show the strongest impact on export growth in the short run. Overall, the findings highlight that Sri Lanka's export performance is shaped more by import dynamics than by exchange rate adjustments and crude oil prices. For policymakers, this means that strengthening supply chains, reducing reliance on exchange rate policies, and enhancing competitiveness will be essential for achieving sustainable export growth.

6.3 Limitations of the Study

While this study provides valuable insights into the short-run determinants of Sri Lanka's export performance, several limitations should be acknowledged to properly interpret the findings and guide future research.

First, the model is limited to three explanatory variables: imports, exchange rate, and crude oil prices. Although these factors are relevant, the model's explanatory power (adjusted R^2 = 14.17%) suggests that a large portion of variation in export growth remains unexplained. Other important influences, such as global demand conditions, foreign direct investment (FDI), inflation, interest rates, political stability, trade agreements, and domestic industrial capacity, were not included due to data and scope constraints. Therefore, the study captures only part of the complex dynamics affecting export performance.

Second, the analysis focuses exclusively on short-run relationships by using the first differences of log-transformed variables. While this approach ensures stationarity and allows interpretation in terms of growth rates, it does not reveal long-term equilibrium relationships. As a result, the findings primarily reflect short-term fluctuations rather than the structural or long-term drivers of trade performance. Future studies could address this gap by employing methods such as cointegration analysis or Vector Error Correction Models (VECM).

Third, the study relies on secondary data from official sources such as the Central Bank of Sri Lanka, the Federal Reserve Bank of St. Louis, and the World Bank. Although these sources are reliable, potential issues such as data revisions, reporting lags, or measurement errors may have influenced the results. Moreover, using aggregate monthly data does not account for sector-specific variations within exports (e.g., agricultural, industrial, or mineral exports), which might respond differently to the studied variables.

Fourth, external shocks such as global financial crises, the COVID-19 pandemic, or geopolitical disruptions were not explicitly considered, even though they could have substantially affected both imports and exports during the study period. Ignoring such structural breaks may limit the generalizability of the results across different time frames.

Finally, the study assumes linear relationships between variables, while trade dynamics may involve non-linear or asymmetric effects. For instance, exchange rate depreciation might benefit some export sectors but harm others, depending on their reliance on imported inputs. In summary, while this study offers meaningful insights into the short-run drivers of export growth, these limitations emphasize the need for future research to expand the scope of variables, explore long-run dynamics, and incorporate sectoral and external shock analyses for a more comprehensive understanding of Sri Lanka's export performance.

6.4 Suggestions

The findings of this study provide several important lessons for policymakers, businesses, and researchers who are concerned with strengthening Sri Lanka's export sector. While the results highlight the influence of imports, crude oil prices, and the exchange rate on export growth, they also point to broader policy directions that the country should consider.

To begin with, the strong and positive link between imports and exports suggests that Sri Lanka's export industry is heavily dependent on imported intermediate and investment goods. This means that policies designed to restrict imports may unintentionally harm export competitiveness. Instead of broad restrictions, policymakers should focus on ensuring that essential imports flow smoothly while discouraging unnecessary luxury imports that drain foreign reserves. A practical approach would be to adopt a strategy of "quality imports for quality exports," where the priority is given to imports that directly support production and innovation. The results show that oil price movements have an insignificant positive effect on exports. Although insignificant, it underlines Sri Lanka's vulnerability to external shocks. This emphasizes the need for diversification. Relying too much on a narrow set of exports leaves the country exposed to global volatility. Developing new industries such as technology-based exports, renewable energy products, and knowledge-driven services can help reduce this dependence and make the export sector more resilient. Although the exchange rate was found to have a negative but statistically insignificant effect on exports, it should not be ignored. The findings indicate that currency depreciation alone is not a reliable tool for promoting exports. Instead, Sri Lanka needs to focus on improving supply-side conditions. Investing in technology, enhancing production efficiency, upgrading value-added industries, and promoting environmentally sustainable practices would provide more durable and meaningful improvements in export competitiveness. Finally, it is important to recognize that the model explained only about 15 percent of the variation in export growth. This means that other crucial factors are at play. Future research should therefore examine a wider range of variables such as global demand, foreign direct investment, domestic infrastructure, logistics efficiency, trade agreements, and policy stability. These elements could provide a more complete picture of what drives Sri Lanka's export performance. In short, the way forward is clear: manage imports carefully, prepare for global shocks, strengthen the supply side of the economy, and broaden the export base. With these measures, Sri Lanka can build a more competitive and sustainable export sector that supports long-term economic growth.

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